

COVID-19 Disease Forecasting: A Spatio-Temporal Analysis of India

Introduction

Infectious disease outbreaks are shaped as much by geography and human behavior as by the pathogen itself — case counts rise and fall in waves, and how quickly an outbreak spreads often depends on population density, mobility patterns, and regional public health capacity. India's COVID-19 experience illustrated this clearly: distinct nationwide waves unfolded alongside sharp state-to-state variation in severity and timing. Understanding these patterns at a sub-national level, rather than relying on national aggregates alone, is essential for meaningful public health planning and resource allocation.

Objective and Approach

This study builds a spatio-temporal forecasting pipeline for COVID-19 case rates at the Indian state-week level, integrating four public data sources — case surveillance data, Google mobility reports, socio-economic indicators, and Census demographic data — into a unified dataset. Engineered features (case-history lags, mobility lags, and a spatial-lag term capturing neighbouring-state activity) feed a gradient-boosted forecasting model, which is evaluated against naive and classical time-series baselines using a time-based train/test split.

Data and Methodology

- COVID-19 was selected over dengue and malaria based on **data availability and integration feasibility**. COVID-19 provides clean daily state-level case data that aligns with Google Mobility data, making it more suitable for reproducible analysis within the project timeframe.
- Four sources were integrated: **COVID-19 case data, Google Mobility data, state-level socio-economic indicators, and Census 2011 demographic data**. Since mobility and socio-economic data were primarily available at the state level, **state-week** was chosen as the common analysis resolution. This avoided replicating state-level values across districts and creating misleading results. Case counts were normalized to **cases per 100,000 population** using Census population data, enabling meaningful comparisons between states of different population sizes.
- **Predictive modelling:** The normalized weekly case rate was used as the target variable. Temporal, mobility, spatial, demographic, and socio-economic features were engineered and used to train a **pooled XGBoost regression model**, with Naive Persistence and per-state ARIMA(1,1,1) used as baseline forecasting approaches.
- **AI-Assisted Development** - AI tools were used as a productivity aid during development, helping reduce repetitive coding and accelerate implementation of data-loading, cleaning, feature-engineering, and modelling scripts. All generated suggestions were reviewed, modified where necessary, and validated against the

actual data and project requirements. The analytical decisions, methodology, feature selection, model choice, evaluation, and interpretation remained the author's responsibility, and all forecasts were generated by the implemented ML/statistical models rather than by AI.

Feature Engineering

Features were engineered to capture **temporal, mobility, and spatial patterns** in COVID-19 spread. These included 1–3 week lags of case counts and case rates, a one-week growth rate, and a three-week rolling case mean. All six mobility indicators were lagged by 1–2 weeks to capture delayed relationships with case rates. A spatial-lag feature was created using neighboring states' previous-week average case rate. **Moran's I** was calculated separately as a spatial autocorrelation diagnostic to assess geographic clustering.

Model Evaluation and Design

Three forecasting approaches were compared: **Naive Persistence**, which uses the previous week's case rate as the next prediction; **per-state ARIMA(1,1,1)** as a classical time-series baseline; and a **pooled XGBoost regressor** as the main model. XGBoost was pooled across states because each state has only about 137 weeks of data, which is insufficient for reliably training separate models, while pooled learning allows the model to learn patterns across states using state-specific features.

A **time-based split** was used, with the final 8 weeks held out for testing, preventing future information from entering training. Model performance was evaluated using **MAE, RMSE, and MAPE**, both overall and separately for **high- and low-burden states**. This provides a more robust assessment by showing whether a model performs well not only on average, but also across states experiencing different levels of disease burden.

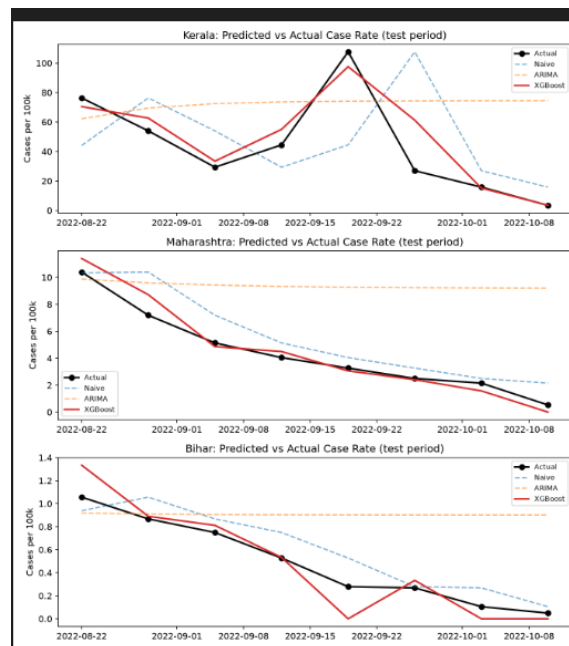
Results

The three forecasting approaches were evaluated on the held-out test period using **MAE, RMSE, and MAPE**. As shown in Table , the **pooled XGBoost model achieved the best overall performance**, with an MAE of **1.273**, RMSE of **5.395**, and MAPE of **99.33%**. Naive Persistence achieved an MAE of **3.738**, RMSE of **10.303**, and MAPE of **165.46%**, while ARIMA(1,1,1) per state recorded higher errors of **9.775**, **17.914**, and **673.91%**, respectively.

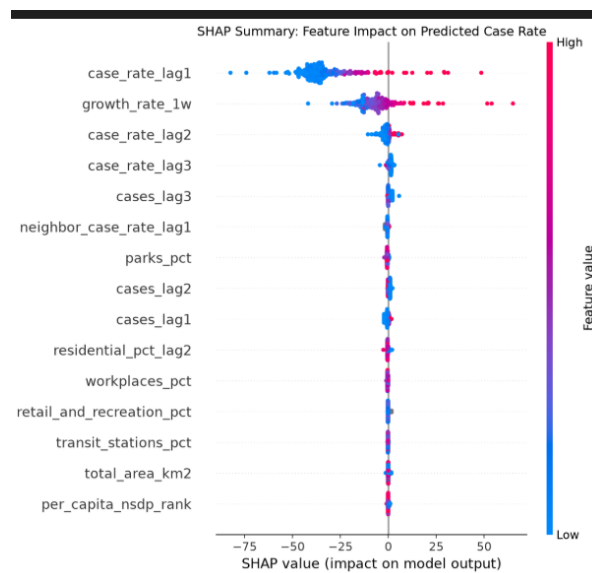
Model	MAE	RMSE	MAPE (%)
Naive Persistence	3.74	10.30	165.5
ARIMA(1,1,1) per state	9.78	17.91	673.9
XGBoost (pooled, full features)	1.27	5.40	99.3

XGBoost also performed better on **high-burden states**, achieving an MAE of **2.03** compared with **7.03** for Naive Persistence and **17.50** for ARIMA. The predicted-versus-actual

trajectories further show that XGBoost was able to capture Kerala's mid-September 2022 case spike that the baseline models missed.



Interpretability: SHAP analysis identified `case_rate_lag1` and `growth_rate_1w` as the dominant predictors. Lagged mobility variables, particularly transit, parks, and workplace mobility, along with the neighbouring-state case-rate feature, also provided meaningful predictive information.



Evaluation caveat: ARIMA was evaluated using a true **8-week-ahead forecast**, without access to actual observations during the test period, whereas Naive Persistence and XGBoost effectively used updated historical values at each test step. Therefore, ARIMA's errors may be inflated relative to a fully equivalent rolling one-step-ahead evaluation.

Limitation and assumptions

- **State-level resolution:** State-week analysis was used due to state-level availability of mobility and socio-economic data.
- **Ladakh:** All six mobility indicators are missing due to the absence of Google Mobility coverage.
- **Telangana:** Census variables are unavailable because the 2011 Census predates its creation.
- **Sikkim:** Growth rate, sex ratio, and literacy are missing due to Census district-name mismatches.
- **Socio-economic data:** Seven states/UTs lack socio-economic ranks because the source covers only 29 of 36.
- **Data vintage:** Socio-economic indicators are older relative ranks, while demographic data is based on the 2011 Census.
- **Missing values:** No imputation was performed; XGBoost handled missing values natively.
- **Model evaluation:** ARIMA used an 8-week multi-step forecast, making its comparison with the effectively one-step-ahead models imperfect.

Missing values were treated as source-level limitations rather than grounds for dropping columns or states. They were retained where possible and handled natively by XGBoost without introducing synthetic values.

Conclusion and Future work

This project developed a reproducible state-week COVID-19 forecasting pipeline integrating case, mobility, demographic, socio-economic, and spatial data. The pooled XGBoost model outperformed the baseline approaches and effectively captured high-burden case patterns. Future work could extend the analysis to district-level modeling, improve baseline evaluation using rolling ARIMA, and further automate the pipeline using n8n to schedule data ingestion, execute the Python pipeline, update model outputs, and generate reports when new data becomes available.

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